F79SP - Theorem List

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1 Asymptotics for the probability to return to zero for Random Walks

Consider simple random walk, with $S_0 = 0$. Then, we compare the probability $P(S_{2n} = 0)$ for both symmetric and non-symmetric cases. Firstly, for the case of symmetric probabilities ($p = q = \frac{1}{2}$). We have that

$$P(S_{2n} = 0) = \binom{2n}{n} \frac{1}{2^{2n}} = \frac{(2n)!}{(n!)^2} \frac{1}{2^{2n}}$$

And by the Stirling's approximation, which states that as $n \to \infty$

$$n! \approx e^{-n} n^n \sqrt{2\pi n}$$

we say that as $n \to \infty$, we have the approximation of

$$P(S_{2n} = 0) \approx \frac{e^{2n}(2n)^{2n}\sqrt{4\pi n}}{e^{-2n}n^{2n}(2\pi n)} \frac{1}{2^{2n}} = \frac{1}{\sqrt{\pi n}} \propto \frac{1}{\sqrt{n}}$$

This probability declines in proportion to $\frac{1}{\sqrt{n}}$. This can be justified by the fact that when n is large, the distribution of S_n is approximately normal with mean 0 and variance n hence standard deviation $\sqrt{n}$.
Then, for the case where $p \neq q$, we continue with

$$\begin{aligned} P(S_{2n} = 0) = \binom{2n}{n} p^n q^n &= \binom{2n}{n} \frac{1}{2^{2n}} (4pq)^n \\ &\approx \frac{e^{2n}(2n)^{2n}\sqrt{4\pi n}}{e^{-2n}n^{2n}(2\pi n)} \frac{1}{2^{2n}} = \frac{1}{\sqrt{\pi n}} \propto \frac{1}{\sqrt{n}} (4pq)^n \\ &= \frac{1}{\sqrt{\pi n}} (4pq)^n \end{aligned}$$

Note that in this case the probabilities decay exponentially fast

Theorem 3.2.1

First we define $\alpha_{b,a} = P($ process S_n reaches state 0 before state b — $S_0 = a)$

$$\alpha_{b,a} = \begin{cases} \dfrac{\left(\dfrac{q}{p}\right)^a - \left(\dfrac{q}{p}\right)^b}{1 - \left(\dfrac{q}{p}\right)^b} & \text{if } p \neq q \\ 1 - \dfrac{a}{b} & \text{if } p = q = \dfrac{1}{2} \end{cases}$$

Proof

Outline of proof:

1. Condition on the first step of the walk to create a relationship between $\alpha_{b,a-1}, \alpha_{b,a}, \alpha_{b,a+1}, \ldots$
2. Use the recurrence relationship to relate the difference of two terms back to $\alpha_{b,0}$ and $\alpha_{b,1}$, since we know that $\alpha_{b,0} = 1$
3. Use the method of difference $(a - b) = (a - c) + (c - d)$ to generate a geometric series, which simplifies calculation.
4. Since we consider b to be fixed and a varying, we can leverage on $\alpha_{b,b} = 0$ to simplify calculations.

We begin by using the Law of Total Probabilities,

$$\begin{aligned} \alpha_{b,a} &= pP(S_n\text{reaches 0 before state b } |S_1 = a+1, S_0 = a) \\ &\quad + qP(S_n\text{reaches 0 before state b } |S_1 = a-1, S_0 = a) = p\alpha_{b,a} + q\alpha_{b,a} \\ &= pP(S_n\text{reaches 0 before state b } |S_1 = a+1) \\ &\quad + qP(S_n\text{reaches 0 before state b } |S_1 = a-1) \quad \textit{(by Markov Property)} \\ &= pP(S_n\text{reaches 0 before state b } |S_0 = a+1) \\ &\quad + qP(S_n\text{reaches 0 before state b } |S_0 = a-1) \quad \textit{(by time homogeneity)} \\ &= p\alpha_{b,a+1} + q\alpha_{b,a-1} \end{aligned}$$

From here we can see that

$$\alpha_{b,a} = p\alpha_{b,a} + q\alpha_{b,a} = p\alpha_{b,a+1} + q\alpha_{b,a-1}$$

$$\begin{aligned}
\implies \alpha_{b,a} - \alpha_{b,a+1} &= \frac{q}{p}(\alpha_{b,a-1} + \alpha_{b,a}) \\
&= \left(\frac{p}{q}\right)^2 (\alpha_{b,a-2} - \alpha_{b,a-1}) \\
&\vdots \\
&= \left(\frac{p}{q}\right)^a (\alpha_{b,0} - \alpha_{b,1})
\end{aligned}$$

We follow that with using the method of difference to find

$$\begin{aligned}
\alpha_{b,0} - \alpha_{b,a} &= (\alpha_{b,a-1} - \alpha_{b,a}) + (\alpha_{b,a-1} - \alpha_{b,a}) + \ldots + (\alpha_{b,0} - \alpha_{b,1}) \\
&= \left(\left(\frac{q}{p}\right)^{a-1} + \left(\frac{q}{p}\right)^{a-2} + \ldots + 1\right)(1 - \alpha_{b,1})
\end{aligned}$$

Case 1: $p = q = \frac{1}{2}$. We arrive at $1-\alpha_{b,a} = a(1-\alpha_{b,1})$. Consider the case where $a = b$ .ie. substitute a with b, we arrive at the conclusion $\alpha_{b,1} = 1 - \frac{1}{b} \implies \alpha_{b,a} = 1 - \frac{a}{b}$ as required.

Case 2: $p \neq q$. We continue that

$$1 - \alpha_{b,a} = \frac{1 - \left(\frac{q}{p}\right)^a}{1 - \frac{q}{p}}(1 - \alpha_{b,1})$$

Letting $b = a$ and $\alpha_{b,b} = 0$ once again, we have

$$\begin{aligned}
1 &= \frac{1 - \left(\frac{q}{p}\right)^b}{1 - \frac{q}{p}}(1 - \alpha_{b,1}) \\
\implies 1 - \alpha_{b,1} &= \frac{1 - \frac{q}{p}}{1 - \left(\frac{q}{p}\right)^b} \\
\implies 1 - \alpha_{b,a} &= \frac{1 - \left(\frac{q}{p}\right)^a}{1 - \frac{q}{p}} \cdot \frac{1 - \frac{q}{p}}{1 - \left(\frac{q}{p}\right)^b} \\
\implies \alpha_{a,b} &= 1 - \frac{1 - \left(\frac{q}{p}\right)^a}{1 - \left(\frac{q}{p}\right)^b} = \frac{\left(\frac{q}{p}\right)^a - \left(\frac{q}{p}\right)^b}{1 - \left(\frac{q}{p}\right)^b}
\end{aligned}$$

as required. From here, we can also deduce the general probability of ruin by setting $b \to \infty$. Hence,

$$P(R) = \alpha_a = \begin{cases} \left(\frac{q}{p}\right)^a & \text{if } p < \frac{1}{2} \\ 1 & \text{if } p \geq \frac{1}{2} \end{cases} \tag{1}$$

Theorem 3.2.3

Suppose $p < \frac{1}{2}$ and $a > 0$. Also, we denote T_a to be the first time S_n reaches state 0. Then,

$$E[T_a] = \frac{a}{q-p} = \frac{a}{1-2p}$$

Proof

Outline of Proof:

1. Acknowledge that in order to reach state 0 from state a, one must first reach all states $a-1, a-2, \ldots, 1, 0$.
2. By time homegeneity, recognize expected time to reach from any state i to state $i-1$ is identical.
3. Form a geometric series and solve for the infinite sum of the series

Let $T_{i,i-1}$ denote the first time the process S_n first reaches state $i-1$ and T_i carries the same meaning as $T_{i,0}$. From there we deduce

$$\begin{aligned} & T_a \equiv T_{a,a-1} + T_{a-1,a-2} + \ldots + T_{1,0} \\ \implies & E[T_a] = a \cdot E[T_1] \end{aligned}$$

Next, we take a closer look on $E[T_1]$ to show that it can be rewritten as

$$\begin{aligned} E[T_1] &= 1 + pE[T_1|\xi_1 = 1] + qE[T_1|\xi_1 = -1] \\ &= 1 + p(1 + E[T_2]) + q(0) \\ &= 1 + pE[T_2] = 1 + 2pE[T_1] \end{aligned}$$

Combining the two concepts we have

$$\begin{aligned} E[T_a] &= a \cdot E[T_1] \\ &= a(1 + 2pE[T_1]) \\ &= a(1 + 2p(1 + 2pE[T_1])) \\ &\vdots \\ &= a + a(2p) + a(2p)^2 + a(2p)^3 + \ldots \\ &= \frac{a}{1-2p} = \frac{a}{q-p} \end{aligned}$$

Chapman Kolmogorov's Equations for Markov Chains

Consider the Markov chain on the state space S. Then, for any $m, n > 0$ and all $i.j \in S$, the Chapman Kolmogorov's equations gives

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}$$

and from there, naturally we have the Chapman-Kolmogorov's inequality that for any single k,

$$p_{ij}^{(m+n)} = p_{ik}^{(m)} p_{kj}^{(n)}$$

Proof

Outline of Proof:

1. Using the Law of Total Probability for the intermediate step
2. Express the probability of the $(n+m)$ step conditioning on the mth step
3. Remove initial condition by the Markov Property

Starting from the definitions,

$$\begin{aligned}
p_{ij}^{(m+n)} &= P(X_{m+n} = j | X_0 = i) \\
&= \sum_{k \in S} P(X_n = k, X_{m+n} = j | X_0 = i) \\
&= \sum_{k \in S} P(X_m = k | X_0 = i) P(X_{m+n} = j | X_m = k, X_0 = i) \\
&= \sum_{k \in S} P(X_m = k | X_0 = i) P(X_{m+n} = j | X_m = k) \quad \textit{by Markov Property} \\
&= \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}
\end{aligned}$$

Theorem 4.2.1

Here, we denote α^C as the vector of absorption probability into closed class C.
1. Vector α^C satisfies:

$$\begin{aligned}
&\alpha^C = P\alpha^C \\
&\alpha^C = 1 \ \ \forall i \in C \\
&\alpha^C = 0 \ \ \forall i \text{ in other closed classes}
\end{aligned}$$

2. $\alpha^C$is the minimum solution of the above system of equations.
3. When the set of states in non-closed classes is finite (in particular when $\mathcal{S}$ is finite) the vector α^C is the unique solution of the above system of equations. Further in this case, the sum of absorption probabilities over all closed classes is 1.

Proof

Here, we only show the proof for (1).
Outline of Proof:

1. For the first equation, we condition on the probability of the first step of the chain

2. Justifications for second and third equation are basically transparent.

For (1), we show that

$$\begin{aligned}
\alpha_i^C &= P_i(X_n\text{ever hits state } C) \\
&= \sum_{j\in\mathcal{C}} p_{ij} P_j(X_n\text{ever hits state } C) \\
&= \sum_{j\in\mathcal{C}} p_{ij}\alpha_j^C \\
&= P\alpha^C
\end{aligned}$$

Then clearly, if $i \in \mathcal{C}$ then $\alpha_i^C = 1$ and $\alpha_i^C = 0$ if i is in other closed classes.

Theorem 4.3.1

For any state $i \in \mathcal{S}$, we have

$$
\begin{aligned}
i \text{ is recurrent} &\Leftrightarrow F_{ii} = 1 \\
&\Leftrightarrow P_i(\exists n > 0 \; s.t. \; X_n = i) = 1 \\
&\Leftrightarrow P(N = \infty) = 1 \\
&\Leftrightarrow E_i(N) = \infty \\
&\Leftrightarrow \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty
\end{aligned}
$$

$$
\begin{aligned}
i \text{ is transient} &\Leftrightarrow F_{ii} < 1 \\
&\Leftrightarrow P_i(\exists n > 0 \; s.t. \; X_n = i) < 1 \\
&\Leftrightarrow P(N = \infty) = 01 \\
&\Leftrightarrow E_i(N) < \infty \\
&\Leftrightarrow \sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty
\end{aligned}
$$

Proof

These sufficient and neccessary conditions are intuitive and should not need proving :).

Theorem 4.3.3

In any class, either all states in the class is transient, or all class is recurrent.
Proof: Outline of proof:
1. Suppose $i, j \in \mathcal{C}$ and we show that i is recurrent $\implies$ j is recurrent.
2. From there, the contrapositive gives us j is transient $\implies$ i is transient.
3. Then, since i, j are arbitrary in the class $\mathcal{C}$, this conclusion can be extended to all elements in the class.

Firstly, note that since $i, j \in \mathcal{C}$, $\exists\, m, n > 0$ such that $p_{ij}^{(m)}, p_{ji}^{(n)} > 0$ respectively. Then suppose i is recurrent, then $\sum_{k=0}^{\infty} p_{ii}^{(k)} = \infty$. From there, we write

$$\begin{aligned}
\sum_{k=0}^{\infty} p_{jj}^{(k)} &\geq \sum_{k=m+n}^{\infty} p_{jj}^{(k)} \\
&= \sum_{k=0}^{\infty} p_{jj}^{(k+m+n)} \\
&\geq \sum_{k=0}^{\infty} p_{ji}^{(n)} p_{ii}^{(k)} p_{ij}^{(m)} \qquad \textit{(by the Chapman-Kolmogorov's inequality)} \\
&= p_{ij}^{(m)} p_{ji}^{(n)} \sum_{k=0}^{\infty} p_{ii}^{(k)} \\
&= \infty
\end{aligned}$$

which shows that j is also recurrent, as required. We further extend this idea to say that in a recurrent class, either all states are **positive recurrent**, or all states are **null recurrent**.

Theorem 4.3.4

Suppose the class $\mathcal{C}$ is recurrent. Then we say that $\mathcal{C}$ is closed. Furthermore, $F_{ij} = 1 \; \forall \; i, j \in \mathcal{C}$.

Proof

Outline of Proof

1. Suppose chain starts in state $j \in C$, then since C is recurrent, return to state j infinitely many times in certain.
2. For some state $i \in S$, we show that if $j \to i$, then $F_{ij} = 1$, which means that i is also in class C.
3. Then since i is chosen to be arbitrary, this means all states that j can reach returns to j certainly, which means the class C is closed.

Following the outline, suppose $j \in C$ and $i \in S$ such that $j \to i$. Then since C is recurrent, $F_{jj} = 1$. From there, we derive

$$\begin{aligned}
1 = F_j j &= P_j(\text{process returns to state j infinitely often}) \\
&= P_j(\text{ process hits state i at some time and hits state j infinitely often}) \\
&\quad + P_j(\text{ process never hits state i and hits state j infinitely often}) \\
&\leq P_j(\text{ process hits state i at some time and hits state j some other time}) \\
&\quad + P_j(\text{ process never hits state i}) \\
&= F_{ji}F_{ij} + (1 - F_{ij}) \\
\implies 0 &\leq F_{ij}(F_{ij} - 1)
\end{aligned}$$

Then, $j \to i \implies F_{ji} > 0 \implies F_{ij} \leq 1$. Note that F_{ij} is a probability and hence $F_{ij} = 1$.

Theorem 4.3.5

If class $\mathcal{C}$ is finite and closed, then it is recurrent.

Proof

Outline of Proof:

1. Note that there must be some $j \in C$ such that i hits state j infinitely often.
2. Using that, we consider for a single case of j and show that it is recurrent
3. Then, since a state in a closed class in recurrent, all states in the class must be recurrent too.

Since $\mathcal{C}$ is finite and closed, then

$$\sum_{j \in \mathcal{C}} P_i(\text{process hits state } j \text{ infinitely often}) = 1$$

Then, for some arbitrary j, we have

$$\begin{aligned}
0 &< P_i(\text{process hits state } j \text{ infinitely often}) \\
&= F_{ij}P_i(\text{process hits state } j \text{ infinitely often}) \\
\implies 0 &< P_j(\text{process hits state } j \text{ infinitely often}) \\
&= P_j(N = \infty)
\end{aligned}$$

Then by Theorem 4.3.1, j is recurrent and thus the whole class $\mathcal{C}$ is too.

Theorem 4.4.4

Suppose Markov chain $\{X_n\}_{n\geq 0}$ such that $\mathcal{S}$ is a finite, single closed class and is aperiodic. Then, its limiting distribution, denoted ω, defined to be

$$\lim_{n\to\infty} p_{ij}^{(n)} = \omega_j$$

would be the unique stationary distribution of the chain.

Proof

Outline of Proof

1. This proof leverages on the definition, which includes the idea of taking $n \to \infty$. We would use this for all the next 3 terms.
2. First we show that ω is indeed a distribution.
3. Then, we show that it is stationary.
4. Lastly, we show that the limiting distribution must be the unique distribution.

We first show that ω is indeed a distribution. We start with saying for all i, n, we have $\sum_{j\in\mathcal{S}} p_{i,j}^{(n)} = 1$ Then, we let $n \to \infty$, and from there

$$\lim_{n\to\infty} \sum_{j\in\mathcal{S}} p_{ij}^{(n)} = \sum_{j\in\mathcal{S}} \omega_j = 1$$

Hence, we shown ω is a distribution as required. Then, for all $n > 0$ and for all $i, j \in \mathcal{S}$, we can show that

$$p_{ij}^{(n+1)} = \sum_{k\in\mathcal{S}} p_{ik}^{(n)} p_{jk}$$

Letting $n \to \infty$, we arrive at

$$\omega_{ij} = \sum_{k\in\mathcal{S}} \omega_k p_{kj}$$

which shows that ω is stationary. as required. We can further investigate to show the uniqueness of the stationary distribution. Suppose π is an arbitrary stationary distribution for our Markov chain such that

$$\pi_j = \sum_{i\in\mathcal{S}} \pi_i p_{ij}^{(n)}$$

Then, letting $n \to \infty$ again, we obtain

$$\pi_j = \sum_{i\in\mathcal{S}} \pi_i \omega_j = \omega_j$$

as required.

Property $\Pi 1$

If $N_1 \sim \Pi(\lambda_1), N_2 \sim \Pi(\lambda_2)$ are independent, then $N_1 + N_2 \sim \Pi(\lambda_1 + \lambda_2)$.

Proof

Outline of Proof:

1. Essentially, we show that the pdf of $N_1 + N_2$ resembles the Poisson distribution, i.e.
2. We split the proof into 2 cases: $N_1 + N_2 = 0, N_1 + N_2 > 0$
3. First case is straightforward and can be derived directly.
4. For $N_1 + N_2 > 0$, we use the fact that given $a + b = k$, then providing the value of one term is sufficient for both terms.
5. Also, note the application of the Newton's binomial, which is the summation series of $(a + b)^k$.

For $k = 0$,

$$P(N_1 + N_2 = k) = P(N_1 = 0, N_2 = 0) = P(N_1 = 0) \cdot P(N_2 = 0) = e^{-\lambda_1 - \lambda_2}$$

For $k > 0$, we have that

$$\begin{aligned}
P(N_1 + N_2 = k) &= \sum_{j=0}^{k} P(N_1 = j, N_2 = k - j) \\
&= \sum_{j=0}^{k} P(N_1 = j) \cdot P(N_2 = k - j) \\
&= \sum_{j=0}^{k} \frac{\lambda_1^j}{j!} e^{-\lambda_1} \cdot \frac{\lambda_2^{k-j}}{(k-j)!} e^{-\lambda_2} \\
&= \frac{e^{-\lambda_1 - \lambda_2}}{k!} \sum_{j=0}^{k} \frac{k!}{j!(k-j)!} \lambda_1^j \lambda_2^{k-j} \\
&= \frac{e^{-\lambda_1 - \lambda_2}}{k!} (\lambda_1 + \lambda_2)^k \\
\implies N_1 + N_2 &\sim \Pi(\lambda_1 + \lambda_2)
\end{aligned}$$

Property $\Pi 2$

Suppose $N_1 \sim \Pi(\lambda_1), N_2 \sim \Pi(\lambda_2)$ are independent, then for any $k = 0, 1, 2, \ldots$ and any $j = 0, 1, \ldots, k$,

$$P(N_1 = j | N_1 + N_2 = k) = \frac{k!}{j!(k-j)!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^j \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{k-j}$$

In other words, conditionally, $N_1 \sim Bin\left(k, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$.

Proof

Outline of Proof:

1. Expand using the definition of the conditional probability
2. Since all $N_1, N_2, N_1 + N_2$ follows a Poisson distribution, we write the pdf accordingly.
3. Rearrange to form the distribution of a Binomial distribution.

$$\begin{aligned} P(N_1 = j | N_1 + N_2 = k) &= \frac{P(N_1 = j, N_1 + N_2 = k)}{P(N_1 + N_2 = k)} \\ &= \frac{P(N_1 = j)P(N_2 = k - j)}{P(N_1 + N_2 = k)} \\ &= \frac{\lambda_1^j e^{-\lambda_1}}{j!} \frac{\lambda_2^{k-j} e^{-\lambda_2}}{(k-j)!} \frac{k!}{(\lambda_1 + \lambda_2)^k e^{-\lambda_1 - \lambda_2}} \\ &= \frac{k!}{j!(k-j)!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^j \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{k-j} \end{aligned}$$

Property $\Pi 3$

Suppose $N \sim \Pi(\lambda)$ where $\lambda > 0$. Also, we say that $R + G = N$ and $R/N = p, G/N = q = 1 - p$. Then,
(i) $R \sim \Pi(\lambda p)$ (ii) $G \sim \Pi(\lambda q)$ (iii) R and G are independent

Proof

Outline of Proof:

1. Note that R and G are symmetrical. Hence, we only need to show for (i).
2. Recognize that we can condition the number of N on the number of R , since $N \geq R$.

3. From there, solve to show that the pdf of R follows a Poisson distribution.

4. To show independence, we show that the joint density function equals to the product of individual density functions. We do this by noting that $N = G + R$.

For (i), we say that for any $k = 0, 1, 2, \ldots$

$$\begin{aligned}
P(R = k) &= \sum_{n=k}^{\infty} P(R = k, N = n) \\
&= \sum_{n=k}^{\infty} P(R = k | N = n) \cdot P(N = k) \\
&= \sum_{n=k}^{\infty} \frac{n!}{(n-k)!k!} p^k q^{n-k} \frac{\lambda^n e^{-\lambda}}{n!} \\
&= \frac{p^k e^{-\lambda}}{k!} \sum_{n=k}^{\infty} \frac{\lambda^n q^{n-k}}{(n-k)!} \\
&= \frac{(\lambda p)^k e^{-\lambda}}{k!} \sum_{n=k}^{\infty} \frac{\lambda^{n-k} q^{n-k}}{(n-k)!} \\
&= \frac{(\lambda p)^k e^{-\lambda}}{k!} \sum_{m=0}^{\infty} \frac{\lambda^m q^m}{m!} \\
&= \frac{(\lambda p)^k e^{-\lambda}}{k!} e^{\lambda q} \\
&= \frac{(\lambda p)^k e^{-\lambda p}}{k!} \\
\implies R &\sim \Pi(\lambda p)
\end{aligned}$$

For (iii) instead, we show that for all $k, i = 0, 1, 2, \ldots$

$$\begin{aligned}
P(R = k, G = i) &= P(R = k, N = k + i) \\
&= P(R = k | N = k + i) \cdot P(N = k + i) \\
&= \frac{(k+i)!}{i!k!} p^k q^i \frac{e^{-\lambda} \lambda^{k+i}}{(k+i)!} \\
&= \frac{\lambda^k}{k!} e^{-\lambda k} \cdot \frac{\lambda^i}{i!} e^{-\lambda i} \\
&= P(R = k) \cdot P(G = i)
\end{aligned}$$

which means that R and G are independent.

Property E4

Suppose $Y_1 \sim Exp(\alpha_1)$,$Y_2 \sim Exp(\alpha_2)$, and Y_1 and Y_2 are independent, then $min(Y_1, Y_2) \sim Exp(\alpha_1 + \alpha_2)$.

Proof

Outline of Proof:

1. If the minimum of two variables is more than x, then both of them must be more than x.
2. Leverage on the independence of the variables.

$$\begin{aligned} P(min(Y_1, Y_2) > x) &= P(Y_1 > x, Y_2 > x) \\ &= P(Y_2 > x) \cdot P(Y_2 > x) \\ &= e^{-\alpha_1 x - \alpha_2 x} \end{aligned}$$

Furthermore, with similar reasoning we can show that $min(Y_1, Y_2, \ldots, Y_n) \sim Exp(\sum \alpha_i)$.

Property Γ3:

Suppose $S_n = \sum Y_i \sim \Gamma(n, \alpha)$ where Y_i are i.i.d.r.v.'s. We define, for $t > 0$,

$$N(t) = \max\{n : S_n \leq t\}$$

which can be interpret as the count of Y within the time $(0, t]$. From there we further define

$$\psi(t) = S_{N(t)+1} - t$$

which can be thought of as the excess at t. Then, for any t, the following properties hold:
(i) $\psi(t) \sim Exp(\alpha)$
(ii) $N(t) \sim \Pi(\alpha t)$
(iii) $\psi(t)$ and $N(t)$ are independent

Proof

Outline of Proof:

1. For (i) and (ii), we condition the excess at time t on the count of $N(t) = n$, by the Law of Total Probability
2. To do this, we find find the joint probability of $N(t) = n$ and $\psi(t) > x$.
3. From there, we can find both the distribution of $\psi(t)$ and $N(t)$ by finding the marginal probability through the joint density.
4. For (iii), we simply show that the joint density is the product of the individual density functions.

We start with some preliminary calculations, where for any $n \in \mathbb{N} \cup \{0\}$ and $x \geq 0$, we have

$$\begin{aligned} P(N(t) = n, \psi(t) > x) &= P(S_n \leq t, S_n + Y_{n+1} > t + x) \\ \implies P(N(t) = 0, \psi(t) > x) &= P(S_0 + Y_1 > t + x) \\ &= e^{-\alpha(t+x)} \end{aligned}$$

From there, we write that for $n > 0$,

$$\begin{aligned} P(N(t) = n, \psi(t) > x) &= \int_0^t f_{S_n}(z) \cdot P(z + Y_{n+1} > t + x)\, dz \\ &= \int_0^t \frac{\alpha^n}{(n-1)!} z^{n-1} e^{-\alpha z} \cdot e^{-\alpha(t+x-z)}\, dz \\ &= \int_0^t \frac{\alpha^n}{(n-1)!} z^{n-1} \cdot e^{-\alpha(t+x)}\, dz \\ &= e^{-\alpha(t+x)} \int_0^t \frac{z^{n-1}\alpha^n}{(n-1)!}\, dz \\ &= e^{-\alpha(t+x)} \frac{(\alpha t)^n}{n!} \end{aligned}$$

(i)

$$\begin{aligned} P(\psi(t) > x) &= \sum_{n=0}^{\infty} P(N(t) = n, \psi(t) > x) \\ &= \sum_{n=0}^{\infty} e^{-\alpha(t+x)} \frac{(\alpha t)^n}{n!} \\ &= e^{-\alpha(t+x)} \sum_{n=0}^{\infty} \frac{(\alpha t)^n}{n!} \\ &= e^{-\alpha(t+x)} e^{-\alpha t} \\ &= e^{-\alpha x} \end{aligned}$$

(ii)

$$\begin{aligned} P(N(t)=n) &= \int_{x=0}^{\infty} P(N(t)=n, \psi(t)>x)\, dx \\ &= \frac{(\alpha t)^n}{n!} e^{-\alpha t} \int_{x=0}^{\infty} e^{-\alpha x}\, dx \\ &= \frac{(\alpha t)^n}{n!} e^{-\alpha t} \end{aligned}$$

(iii) Following from the results of (i) and (ii), then

$$\begin{aligned} P(N(t)=n, \psi(t)>x) &= \frac{(\alpha t)^n}{n!} e^{e^{-\alpha(t+x)}} \\ &= \frac{(\alpha t)^n}{n!} e^{e^{-\alpha t}} e^{-\alpha x} \\ &= P(N(t)=n) \cdot P(\psi(t)>x) \end{aligned}$$

Property PP1

We start with the definition used to construct the PP. Let $\{Y_i\}$ be i.i.d.r.v having an $Exp(\lambda)$ distribution. Let $S_0 = 0$ and $S_n = \sum Y_i$. For $t \geq 0$, we write

$$N(t) = \max\{n : S_n \leq t\}$$

Then $\{N(t), t \geq 0\}$ is a homogeneous PP. We denote

$$N(t_1, t_2) = N(t_2) - N(t_1)$$

With all that said, the following properties follows:
(i) for all $0 \leq t_1 < t_2$,

$$N(t_1, t_2) \sim \Pi(\lambda(t_2 - t_1))$$

(ii) for any $0 < t_1 < t_2, \ldots, t_l$, the random variables $N(t_1, t_0), N(t_2, t_1), \ldots, N(t_{l-1}, t_l)$ are mutually independent.

Proof

Outline of Proof:

1. We only provide proof for (i) here.
2. Essentially, we show that the pdf of $N(t_1, t_2)$ matches a Poisson distribution with parameters $\lambda(t_2 - t_1)$
3. We split this into two cases: firstly, if $t_1 = 0$, then we can use the property $\Gamma 3$.
4. For $t_1 > 0$, we condition the count of $N(t_2, t_1)$ on the count of $N(0, t_1)$
5. Then, we redefine the waiting times such that Y_1 is the "excess" at t_1, and continue from there.

(i) If $t_1 = 0$, then take $t = t_2$ and from property $\Gamma 3$ from above,

$$N(t_1, t_2) = N(t_2) - N(t_1)$$

Since $N(t_1) \sim \Pi(\lambda t_1)$ and $N(t_2) \sim \Pi(\lambda t_2)$. Then from the basic property of the Poisson distribution we have our desired result. Then, for $t_1 > 0$, for all $k = 0, 1, 2, \ldots$,

$$P(N(t_1, t_2) = k) = \sum_{j=0}^{\infty} P(N(t_1) = j, N(t_1, t_2) = k)$$

Then, by Property $\Gamma 3$ above, we denote $\bar{Y}_1 = \psi(t_1), \bar{Y}_2 = Y_{j+2}, \bar{Y}_3 = Y_{j+3}, \ldots$ and also $\bar{S_n} = \sum \bar{Y}_i$. Then, on the event $\{N(t_1) = j\}$, we can say that

$$N(t_1, t_2) = \max\{n : \bar{S}_n \leq t_1 - t_2\}$$

With that, we can say that given $N(t_1) = j$, ,the random variables $\{\bar{Y}_n\}$ are mutually independent and have $Exp(\lambda)$ distribution. Then, for any $j = 0, 1, 2, \ldots$,

$$\begin{aligned}
P(N(t_1, t_2) = k) &= \sum_{j=0}^{\infty} P(N(t_1) = j, N(t_1, t_2) = k) \\
&= \sum_{j=0}^{\infty} P(N(t_1) = j) \cdot P(N(t_1, t_2) = k | N(t_1) = j) \\
&= \sum_{j=0}^{\infty} P(N(t_1) = j) \cdot \frac{(\lambda(t_2 - t_1))^k}{k!} e^{-\lambda(t_2 - t_1)} \\
&= \frac{(\lambda(t_2 - t_1))^k}{k!} e^{-\lambda(t_2 - t_1)} \\
\implies N(t_1, t_2) &\sim \Pi(\lambda(t_2 - t_1))
\end{aligned}$$

Theorem 5.10.2

Superposition Theorem. Let $N_1(\cdot), \ldots, N_k(\cdot)$ be independent PP with rates $\lambda_1, \ldots, \lambda_k$. Then, we let process $N(\cdot) = \sum_{i=1}^{k} N_i(\cdot)$. be the superposition of these process. Then,

(i) $N(\cdot)$ is a PP with rate $\lambda = \sum_{i=1}^{k} \lambda$
(ii) Given the occurrence of a point of the process $N(\cdot)$ at time t, the probability it belongs to any $N_i(\cdot)$ is $p_i = \lambda_i/\lambda$, independently of all else

Proof

Outline of Proof:

1. For the first part, simply use the property of Poisson distribution that the sum of variables each following a Poisson distribution would also follow a Poisson distribution with the rate being the sum of individual rates.
2. For the second part, use Property $\Pi 2'$ which mentions about the "Binomial distribution" behaviour of the Poisson random variables.

For part (i), since the condition PP1 is satisfied for all the individual processes $N_i(\cdot)$, then by the property of Poisson distribution, for any $0 \leq t_1 < t_2$,

$$N(t_1, t_2) = \sum_{i=1}^{k} N_i(t_1, t_2) \sim \Pi(\lambda(t_2 - t_1))$$

Then, for part (ii), we have that from Property $\Pi 2^{'}$ that given $N_1 \sim \Pi(\lambda_1), N_2 \sim \Pi(\lambda_2), \ldots, N_k \sim \Pi(\lambda_k)$,

$$\begin{aligned} P(N_j = 1 | N = \sum_{i=1}^{k} N_i = 1) &= \binom{1}{1} \left(\frac{\lambda_i}{\lambda}\right)^1 \left(\frac{\lambda_i}{\lambda}\right)^{1-1} \\ &= \frac{\lambda_i}{\lambda} \end{aligned}$$

Theorem 5.10.3

Splitting Theorem. Let $N(\cdot)$ be a PP with rate λ. Then, suppose each point of $N(\cdot)$ is assigned to a new point process $N_i(\cdot)$ where $1 \leq i \leq k$, with probability p_i, $(\sum_{i=1}^{k} p_i = 1)$. Then, we can say that $\{N_i(\cdot)\}_{1 \leq i \leq k}$ are independent PP with rates , $p_i\lambda$,respectively.

Proof

Outline of Proof:

1. Show that the sum of the points in all individual processes are equal to the original process.
2. Use results from Property Π3 and the fact that a point belonging to a certain process N_i is λ_i.

$N(\cdot)$ is a PP with rate λ means that for any $0 \leq t_1 < t_2$, we have

$$N(t_1, t_2) \sim \Pi(\lambda(t_2 - t_1))$$

Then, if we carry out the splitting action as mentioned, obviously the sum of all the points will still be equal to the original, namely

$$\sum_{i=1}^{k} N_i(t_1, t_2) = N(t_1, t_2)$$

From there, using results from Property Π3, we can say that for all $1 \leq i \leq k$,

$$N_i(t_1, t_2) \sim \Pi(\lambda p_i(t_2 - t_1))$$

and all $N_i(t_1, t_2)$ are mutually independent. Hence, since all the individual counting processes fulfill the conditions in the Second Definition of PP, we can say that all of the counting processes are independent PP with respective rates λp_i.

Theorem 5.11.1

The Second and the Third definitions of a PP are equivalent. In other words, given PP3(ii), the conditions PP3(i) and PP1(i) are equivalent. Here are the conditions mentioned:

PP3(ii) : The number of points $N(I_1), \ldots, N(I_k)$ occurring in disjoint finite intervals are mutually independent.
PP3(i) : For any $t > 0$, and small h, we have

$$\begin{aligned} P(N(t, t+h) = 0) &= 1 - \lambda h + o(h) \qquad \text{as } h \to 0 \\ P(N(t, t+h) = 1) &= \lambda h + o(h) \qquad \text{as } h \to 0 \\ P(N(t, t+h) \geq 2) &= o(h) \qquad \text{as } h \to 0 \end{aligned}$$

PP1(i) : For any finite interval $I = [t_1, t_2]$, the random variable $N(I)$ has a Poisson distribution with mean $\int_{t_1}^{t_2} \lambda(t)\, dt$.

Proof

Outline of Proof:

1. For the direction PP3(i) $\implies$ PP1(i), we show that the pdf of $N(I)$ for any finite interval I corresponds to the pdf of the Poisson distribution.
2. This is done through induction.
3. In the induction, we construct a DE and solve for the desired probability.
4. For the direction PP1(i) $\implies$ PP3(i), we simply show the equivalence of the expressions under the condition $h \to 0$.

First, we show that given PP3(ii), the condition PP3(i) $\implies$ PP1(i). Suppose $n = 0, 1, 2, \ldots, t_1$ and $t \geq 0$, we let $p_n(t) = P(N[t_1, t_1 + t] = n)$. Then, we wish to show that for all n and all t,

$$N(t_1, t_1 + t) \sim \Pi(\lambda t)$$

which is required for PP1(i). This is the equivalent of

$$p_n(t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$

and we will show this by induction on n. Firstly, for $n = 0$, by the definition given in PP3(i), for all $h > 0$

$$\begin{aligned} p_0(t + h) &= p_0(t) p_0(h) \\ &= p_0(t)[1 - \lambda h + o(h)] \qquad \text{as } h \to 0 \end{aligned}$$

Then,

$$\frac{p_0(t+h)-p_0(t)}{h} = \frac{p_0(t)[1-\lambda h+o(h)]-p_0(t)}{h} = -\lambda p_0(t) + \frac{o(h)}{h}$$

as $h \to 0$. From there, we see that $p_0'(t) = -\lambda p_0(t)$, and solving the differential equation gives us $p_0(t) = ce^{-\lambda t}$. We know that $p_0(0) = 0 \implies c = 1$, then finally, we arrive at $p_0(t) = e^{-\lambda t}$ as required. Now, suppose that the result holds for the case $n = k-1$, and we show that it holds for the case $n = k$ too. First, we write

$$\begin{aligned} p_k(t+h) &= P(N(0,t)=k)\cdot P(N(t,t+h)=0) + P(N(0,t)=k-1)\cdot P(N(t,t+h)=1) \\ &\quad + P(N(0,t)<k-1)\cdot P(N(t,t+h)>1) \\ &= p_k(t)p_0(h) + p_{k-1}(t)p_1(h) + o(h) \\ &= p_k(t)(1-\lambda h) + p_{k-1}(t)\lambda h + o(h) \end{aligned}$$

Then, with similar flow to the case where $n = 0$, we write that as $h \to \infty$,

$$\begin{aligned} \frac{p_k(t+h)-p_k(t)}{h} &= -\lambda p_k(t) + \lambda p_{k-1} + \frac{o(h)}{h} \\ &= -\lambda p_k(t) + \frac{t^{k-1}\lambda^k}{(k-1)!}e^{-\lambda t} + \frac{o(h)}{h} \end{aligned}$$

From there, we write that

$$\begin{aligned} p_k'(t) &= -\lambda p_k(t) + \frac{t^{k-1}\lambda^k}{(k-1)!}e^{-\lambda t} \\ \implies \frac{t^{k-1}\lambda^k}{(k-1)!} &= e^{\lambda t}p_k'(t) + \lambda e^{\lambda t}p_k(t) \\ &= \frac{d}{dt}(e^{\lambda t}p_k(t)) \\ \implies e^{\lambda t}p_k(t) &= \int \frac{t^{k-1}\lambda^k}{(k-1)!}\,dt \\ &= \frac{(\lambda t)^k}{k!} + c \end{aligned}$$

Then since $p_k(0) = 0 \implies c = 0$, we have

$$e^{-\lambda t}p_k(t) = \frac{(\lambda t)^k}{k!} \implies p_k(t) = \frac{(\lambda t)^k}{k!}e^{-\lambda t}$$

as required. This concludes the part where we show that given PP3(ii), the conditions PP3(i) $\implies$ PP1(i). For the next part, we show the converse. For any $t \geq 0$, as $h \to$,

$$P(N(t,t+h)=0) = e^{-\lambda h} = 1 - \lambda h + o(h)$$

(Note how the two expressions are equivalent as $h \to$). Similarly, we can write

$$P(N(t, t+h) = 0) = \lambda h e^{-\lambda h} = \lambda h + o(h)$$

Then finally, for the complementary event,

$$\sum_{k=0}^{\infty} P(N(t, t+h) = k) = 1 \implies P(N(t, t+h) \geq 2) = o(h)$$

With that, PP3(i) follows.

Lemma 5.13.1

Suppose $N(t)$ represents the process of claim arrivals, and X_n is the size of the nth claim. Further let $W(t)$ be the total of all claims within the time interval of length t. We also know that for an individual X, it is given that $E[X] = a$ and $E[X^2] = b$. Then, The mean and variance of $W(t)$ are given by

$$E[W(t)] = a\lambda t \qquad Var[W(t)] = b\lambda t$$

1.1 Proof

Outline of Proof:

1. For both, we use the Law of Total Probability to condition on all the possible values of $N(t) = n$.

Since PP is time-homogeneous, we consider the interval $(0, t)$. Since $W(t) = \sum_{i=1}^{N(0,t)} X_i$, then we have

$$\begin{aligned}
E[W(t)] &= E\left(\sum_{i=1}^{N(0,t)} X_i\right) \\
&= \sum_{n=1}^{\infty} \left(P(N(t) = n) \cdot \sum_{i=0}^{n} X_i\right) \\
&= \sum_{n=1}^{\infty} \left(P(N(t) = n) \cdot na\right) \\
&= a \sum_{n=1}^{\infty} \left(nP(N(t) = n)\right) \\
&= aE[N(t)] \\
&= a\lambda t
\end{aligned}$$

Then, furthermore, we have

$$
\begin{aligned}
E[W(t)^2] &= E\left(\sum_{i=1}^{N(0,t)} X_i\right)^2 \\
&= \sum_{n=1}^{\infty} P(N(t)=n)\cdot E\left(\sum_{i=0}^{n} X_i\right)^2 \\
&= \sum_{n=1}^{\infty} P(N(t)=n)\cdot (nE[X^2] + n(n-1)(E[X])^2) \\
&= \sum_{n=1}^{\infty} P(N(t)=n)\cdot (nE[X^2] - n(E[X])^2) + n^2 n(E[X])^2) \\
&= \sum_{n=1}^{\infty} P(N(t)=n)\cdot (n\sigma^2 + n^2 a^2) \\
&= \sigma^2 \sum_{n=1}^{\infty} nP(N(t)=n) + a^2 \sum_{n=1}^{\infty} n^2 P(N(t)=n) \\
&= \sigma^2 E[N(t)] + a^2 E[N(t)^2] \\
&= \sigma^2 \lambda t + a^2((\lambda t)^2 + (\lambda t))
\end{aligned}
$$

$$
\begin{aligned}
\implies Var[W(t)] &= E[W(t)^2] - E[W(t)]^2 \\
&= (a^2 + \sigma^2)\lambda t \\
&= b\lambda t
\end{aligned}
$$

Theorem 6.3.1: FDEs and BDEs

These are used to recover probability functions $(p_{ij})_{i,j\in S}$ given the transition rate matrix Q. Formally, we say that:
Given Q, we have the FDE

$$p_{ij}^{'}(t) = \sum_{k\in S} p_{ik}(t)q_{kj}$$

and also the BDE

$$p_{ij}^{'}(t) = \sum_{k\in S} q_{ik}p_{kj}(t)$$

Proof:

Outline of Proof:

1. Start with the Chapman-Kolmogorov's equation for the continuous Markov Process.
2. From there, we obtain a two-part transition probability.
3. Differentiating with respect to each of the parts gives us the FDE and BDE respectively.

With the Chapman-Kolmogorov equations, we have

$$p_{ij}(s+t) = \sum_{k\in S} p_{ik}(s)p_{kj}(t)$$

for all $i, j \in S, s, t > 0$. Then, we differentiate with respect to t , at let $t = 0$ to obtain the FDE,

$$p_{ij}^{'}(s) = \sum_{k\in S} p_{ik}(s)p_{kj}^{'}(0)$$

and differentiate with respect to s and let $s = 0$ to obtain the BDE,

$$p_{ij}^{'}(t) = \sum_{k\in S} p_{ik}^{'}(0)p_{kj}(t)$$

Theorem 6.5.2

A distribution π is stationary iff

$$\pi Q = 0 \quad \text{or equivalently} \quad \sum_{i \in S} \pi_i p_{ij} = 0 \ \ \forall j \in S$$

Proof

Outline of Proof:

1. Start with the original definition of a stationary distribution.
2. We recognize that the equation holds for all $t > 0$, hence let $t = 0$ and we have $p_{ij}(0)' = q_{ij}$.

Suppose S is finite, then for all $j \in S$

$$\begin{aligned} \pi P(t) = \pi \implies & \sum_{i \in S} \pi_i p_{ij}(t) = \pi_j \quad \text{for all } t \geq 0 \\ = & \sum_{i \in S} \pi_i p_{ij}(t)' = 0 \quad \text{for all } t \geq 0 \\ = & \sum_{i \in S} \pi_i q_{ij} = 0 \end{aligned}$$

as required.